Ex.No. 4: OFDMA Resource Block Allocation

Objective:

To determine the **Number of Resource Blocks (RBs)** allocated in an OFDMA system based on available bandwidth and system configuration.

Objective 1: Matlab Code and Output

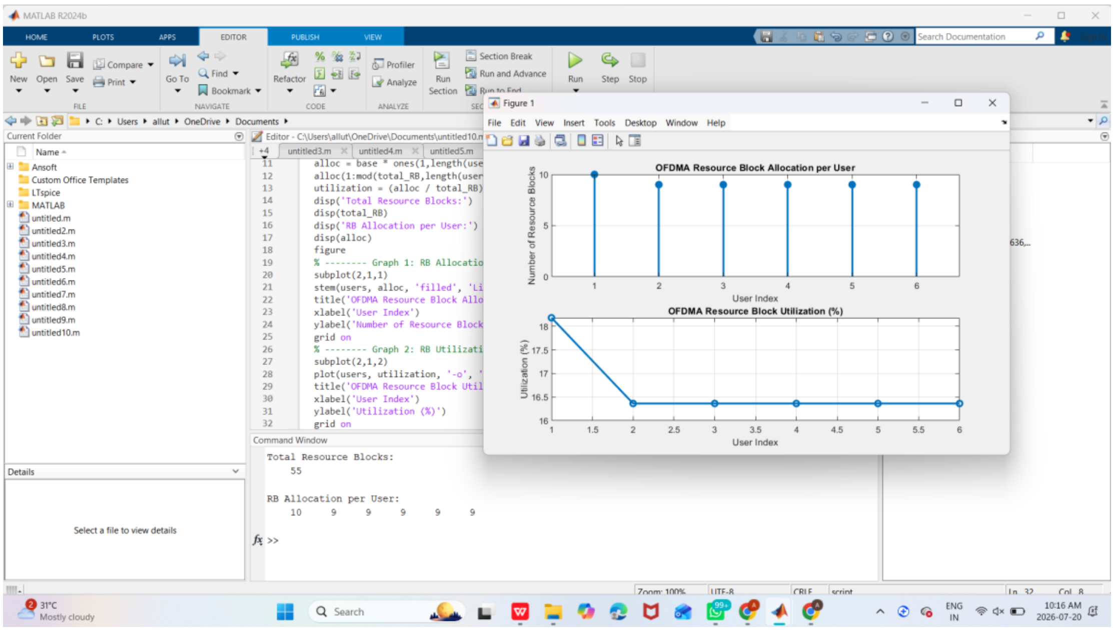
```
clc; clear; close all;

bandwidth = 10e6;
rb_size = 180e3;

total_RB = floor(bandwidth / rb_size);

users = 1:6;
base = floor(total_RB / length(users));alloc = base * ones(1,length(users));
alloc(1:mod(total_RB,length(users))) = alloc(1:mod(total_RB,length(users))) + 1;
utilization = (alloc / total_RB) * 100;
disp('Total Resource Blocks:')
disp(total_RB)
disp('RB Allocation per User:')
disp(alloc)
figure
subplot(2,1,1)
stem(users, alloc, 'filled', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on
subplot(2,1,2)
plot(users, utilization, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on
```

OUTPUT:



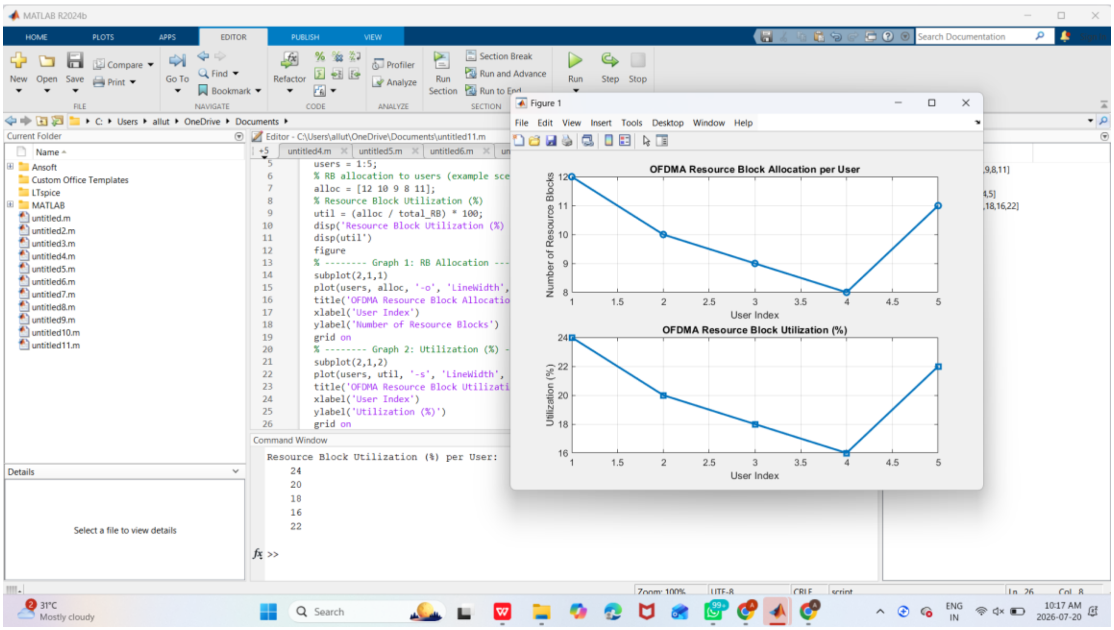
Objective 2:

To evaluate the **Resource Block Utilization (%)** by analyzing how efficiently available RBs are assigned to active users.

Matlab Code and Output :

```
clc; clear; close all;
total_RB = 50;
users = 1:5;
alloc = [12 10 9 8 11];
util = (alloc / total_RB) * 100;
disp('Resource Block Utilization (%) per User:')
disp(util')
figure
subplot(2,1,1)
plot(users, alloc, '-o', 'LineWidth', 2)
title('OFDMA Resource Block Allocation per User')
xlabel('User Index')
ylabel('Number of Resource Blocks')
grid on
subplot(2,1,2)
plot(users, util, '-s', 'LineWidth', 2)
title('OFDMA Resource Block Utilization (%)')
xlabel('User Index')
ylabel('Utilization (%)')
grid on
```

OUTPUT:



Objective 3:

To calculate the **User Throughput (Mbps)** based on allocated resource blocks and modulation efficiency in the OFDMA system.

Objective 3: Matlab Code and Output

```
clc; clear; close all;
rb_bw = 180e3;
users = 1:5;
alloc_RB = [12 10 9 8 11];

mod_eff = [2 4 6 4 2];throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
disp('User Throughput (Mbps):')
disp(throughput')
figure

subplot(2,1,1)plot(users, throughput, '-o', 'LineWidth', 2)
title('OFDMA User Throughput')
```

```
xlabel('User Index')
ylabel('Throughput (Mbps)')
grid on
subplot(2,1,2)
plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
title('Resource Blocks vs Throughput Relationship')
xlabel('Allocated Resource Blocks')
ylabel('Throughput (Mbps)')
grid on
```

OUTPUT: